

Aritra Roy

Computational Materials Science | PhD Researcher

✉ pgr.aritra.roy@lsbu.ac.uk

🌐 www.aritaroy.live

☎ +44 7393062351



PhD researcher developing ComProScanner, a multi-agent text-mining framework for piezoelectric materials discovery. Expertise in LLMs, RAG systems, fine-tuning, AI agents, DFT/VASP calculations, and deep learning for materials informatics. Published researcher with a strong Python programming and computational chemistry background.

EDUCATION

- 2023 – Present ◇ **Ph.D., London South Bank University, UK** in Chemical Engineering
Thesis title: *Text Mining and Deep Learning for Piezoelectric Materials Discovery*.
- 2019 – 2021 ◇ **M.Sc. Chemistry, Pondicherry University, India** in Computational Chemistry
Thesis title: *Finding a More Stable Semiconductor Borophene Using the Theoretical Approach*.
- 2016 – 2019 ◇ **B.Sc. Chemistry, RKM Vivekananda Centenary College, India** in Chemistry Honours

RESEARCH EXPERIENCE

- 2023 – Present ◇ **Doctoral Researcher**, London South Bank University, UK.
• Developed ComProScanner: generalisable multi-agent platform for automated materials composition-property database construction using LLM agents and RAG
• Demonstrated ComProScanner's capability by extracting piezoelectric materials data from 10,000+ scientific papers across 4 major publishers
• Built high-throughput VASP DFT framework performing ~2,000 piezoelectric property calculations on doped ZnO supercells
- 2024 – Present ◇ **Visiting Researcher**, King's College London, UK.
• Cross-institutional collaboration on advanced materials characterisation
• Contributing to computational modelling of piezoelectric system under the supervision of *Dr Chiara Gattinoni*
- 2022 – Present ◇ **Guest Researcher**, Fantuzzi Group, University of Kent, UK.
• Performed DFT calculations supporting experimental works on main-group chemistry and organometallic chemistry
• Resulted in five co-authored publications including one *JACS* with *Dr Felipe Fantuzzi*

RESEARCH PUBLICATIONS

PhD Research

- 1 A. Roy, E. Grisan, J. Buckeridge, and C. Gattinoni, "Comproscanner: A multi-agent based framework for composition-property structured data extraction from scientific literature," *Digital Discovery*, Accepted, 2026. [doi](#) [📄](#) [🔗](#)

Collaborative Research (Selected)

- 1 D. Das et al., "A copper complex receptor for nanomolar sulfide sensing and applications in dna/bsa binding," *Journal of Photochemistry and Photobiology A: Chemistry*, vol. 461, p. 116 154, 2025. [doi](#)
- 2 S. Kamalinahad, A. Roy, P. Gamallo, and F. Fantuzzi, "Engineering high-capacity hydrogen storage in pristine Ca₁₂O₁₂ nanocages via cooperative adsorption," *New Journal of Chemistry*, 2025. [doi](#)

- 3 D. Dhara et al., "A discrete trialane with a near-linear al₃ axis," *Journal of the American Chemical Society*, vol. 146, no. 49, pp. 33 536–33 542, 2024. [doi](#)

Workshop/Hackathon Contribution

- 1 Y. Zimmermann et al., "32 examples of LLM applications in materials science and chemistry: Towards automation, assistants, agents, and accelerated scientific discovery," *Mach. Learn. Sci. Technol.*, vol. 6, no. 3, p. 030 701, 2025. [doi](#) [↗](#)

TECHNICAL SKILLS

Machine Learning & AI	◇ PyTorch, TensorFlow, scikit-learn, LangChain, LangGraph, CrewAI, Hugging Face Transformers, Unsloth AI, NumPy, Pandas, NLP, Data Science, Prompt engineering, Fine-tuning (Gemma, Llama 3), RAG, AI Agents
Programming	◇ Python 3.x, Bash/Shell, C, JavaScript, SQL, HTML/CSS, PHP, $\LaTeX$
Computational Chemistry	◇ DFT calculations, VASP, Gaussian 16, ORCA, VESTA, Phonopy, GaussView, Chemcraft, Multiwfn, SLURM
Development Tools	◇ VS Code, Jupyter Notebook, Git/GitHub, Docker, Anaconda, MongoDB, Postman, Netlify, Hostinger
Web Technologies	◇ React, Astro, Bootstrap, Tailwind CSS, jQuery, Node.js, Laravel

AWARDS & ACHIEVEMENTS

- 2026 ◇ **Alchemy Travel Grant:** Secured £7,500 as a collaborative travel fund from Alchemy to visit Prof. Berend Smit's lab at Swiss Federal Technology Institute of Lausanne (EPFL), Switzerland.
- 2025 ◇ **Winter School: Robotics & AI for Materials Chemistry**, Alchemy, University of Liverpool. Selected as 1 of 30 students internationally for intensive hands-on workshop (Dec 8–12, 2025).
- 2024 ◇ **Top 5 Team – Devpost Hackathon**, MOFMaster Project. Developed AI-powered tool for MOFid generation and CH₄/N₂ gas separation prediction. Fine-tuned Llama-3-8B model and implemented MOF-GRU architecture for property prediction. [↗](#)
- 2021 ◇ **3rd Position in M.Sc. Chemistry**, Pondicherry University (CGPA: 8.88/10).

TECHNICAL BLOG ARTICLES (Selected)

- ◇ AI-ML Powered Materials Design Breakthrough: Is the Colorful Future of Materials Science Near?
- ◇ How to Make Your Chemical Synthesis Absolutely Easier Using AI Advantage
- ◇ Fascinating Power of Googling in Computational Chemistry
- ◇ Introduction To Computational Chemistry Calculations: PES and Saddle Point

LANGUAGES

■ **English:** Fluent (IELTS Academic: 7.0, CEFR Level C1) ■ **Bengali:** Native ■ **Hindi:** Conversational

REFERENCES

- ◇ Available upon request from: Dr John Buckeridge (London South Bank University, UK), Dr Chiara Gattinoni (King's College London, UK), Dr Felipe Fantuzzi (University of Kent, UK).